

Understanding of Modern Software-Defined Networking

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Abstract—This report summarizes an investigation of modern Software-Defined Networking (SDN), focusing on the evolution from the traditional network architecture, where the control plane and data plane are tightly coupled within individual network devices, to early SDN architecture that separate network control from packet forwarding, and ultimately to modern programmable network infrastructures deployed in large-scale data centers. The study examined ONOS, P4, Stratum, P4Runtime and Leaf-Spine Fabric. Particular attention was given to the separation of control and data planes, controller abstractions, programmable forwarding pipelines, and production-scale fabric architectures. The report concludes with some of the use cases, observations on the architectural tradeoffs and research challenges associated with programmable networks.

Index Terms—Computer Network, Software-Defined Network, SDN

I. INTRODUCTION

Started back in 1952, when IBM first manufactured large computer system and sold into the market as mainframe. It was bulky, large, and expensive machine that take up a room size. More importantly, every components within that mainframe is IBM's proprietary hardware, which means, a customer have to buy the whole vertically integrated solution from a single vendor regardless of the solution they want to have for their problem (software calculation, hardware design, finance...).

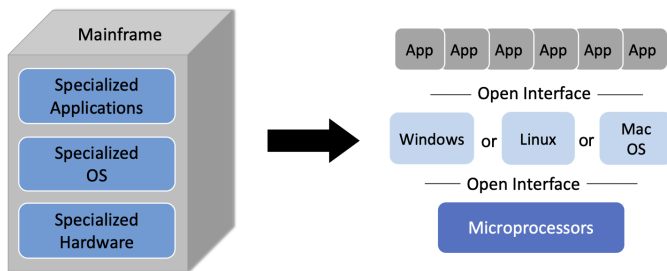


Fig. 1. Transformation from vertical market to horizontal marketplace with open interfaces at all level. Adapted from [1].

The evolution began with the arrival of open-source operating systems and microprocessors, which helped transform the market into a horizontal one where innovation could happen at any layer.

SDN initiatives draw the similar inspiration by decoupling the control plane from the data plane. Decision-making logic is

decided by a external¹ logically centralized controller², using information from network operating systems enabled on top of the underlying switches. This abstraction simplifies switch's role, off-loading complex decision making logic (i.e., routing, path computation, ACL..) off-device and reducing the switch's primary task to packet forwarding. Such devices are therefore referred to as "white box" or "bare-metal" switches³ which does not ship with a fixed network OS, creating a hardware ecosystem for multiple operating system vendors.

OpenFlow is the pioneer protocol that support this disaggregation in the early stage of SDN. Although it was a main instrumental propelling SDN into the mainstream, OpenFlow now play a small role of today SDN because of an architectural limitation it imposed—namely, its fixed-function match-action pipeline. Modern infrastructure has largely transitioned toward fully programmable data planes driven by the P4 programming language, P4Runtime, and Stratum. The objective of this study is to explore this architectural evolution from early OpenFlow to current P4-based ecosystems deployed across large-scale hyperscale data centers. Unless otherwise noted, the architectural exposition in the remainder of this report, including the accompanying figures, follows the treatment given in [1].

II. BASIC ARCHITECTURE

The core philosophy of Software-Defined Networking is horizontal disaggregation across every layers of the networking stack [1]. Rather than relying on a vertically integrated, propriety hardware appliances, SDN decompose them into abstract components that is independent from other components, allowing each of which being used interchangeably and innovated at each layer.

- **Local Network OS layer:** Physical switch hardware is decoupled from proprietary vendor software through

¹In SDN terminology, a distinction exists between **external (global) network OS** and the **internal (local) network OS**. The External NOS refers to a distributed controller platform to maintain a centralized view of the network (ONOS section) while Internal OS refers to lightweight OS running directly on-switch to manage internal switch operation and communicate with the external network OS.

²While SDN control plane is logically centralized to provide a unified global view of network state and simplify network management, production deployment (introduced in ONOS section) are physically distributed for high availability and scalability

³"bare-metal" switches run a ONIE bootloader uses facilities in Linux kernel and BusyBox, defining an open source "install environment" for end users to install target network OS [3]

bare-metal whitebox switches. Utilizing an Open Network Install Environment (ONIE) bootloader [3], the switch can load any compatible local Network OS (e.g., Stratum, SONiC) to handle local hardware initialization and peripheral management.

- **Control-to-Data Communication Interface:** The control plane and data plane interact over well-defined consumer/provider API endpoints using standard protocol (e.g., OpenFlow, gRPC/P4Runtime). This bidirectional interface allows controller to program forwarding rules while collecting real-time telemetry, link states, metrics, and packet from the data plane.
- **Device Driver & Pipeline Layer:** This layer resolves hardware heterogeneity by splitting device control into low-level driver abstractions and pipeline-mapping logic. Analogous to a hardware SDK, the *device driver* handles low-level chip management, translate control protocol messages (e.g., OpenFlow or P4Runtime) directly into chip memory while absorbing vendor-specific behavior mismatches. Sitting above the driver, the *pipeliner* maps generic, high-level *flow objectives* into the exact multi-table forwarding pipeline required by the underlying ASIC, freeing higher-layer application logic from needing to know the switch’s internal table layout.
- **Data Plane layer:** Packet processing logic is completely decoupled from fixed-function hardware ASIC pipeline, allowing enterprise to implement custom header protocols on demand instead of waiting for switch vendor for the protocol implementation in their silicon chips. Using P4 language, packet parsing, header modifications, and forwarding behaviors can be programmed, updated, and recompiled directly on the hardware at runtime.

A. Single-switch perspective: The Software Stack

This includes an overview of a SDN stack with the involvement of a switch and controllers, highlight the top-to-bottom software stack being used in the SDN architecture. The main component include the Control App, Network OS, Switch OS, and P4 program—which used for the programmable switch. Figure 2 also introduced an exemplar of open-source software we used for this report (e.g. SD-Fabric, ONOS, Stratum), and protocol uses for communication between each layer.

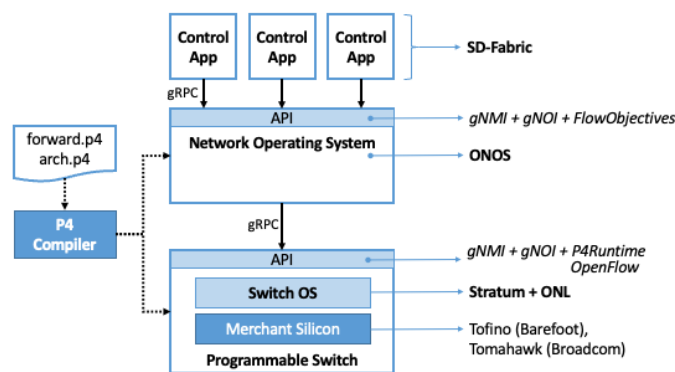


Fig. 2. Overall architecture of SDN stack with one switch. Adapted from [1].

At the top of the stack, Control Applications communicate with the Network OS via gRPC over Northbound APIs. Rather than writing raw table entries, applications issue high-level, topology-agnostic *Intent*, or hardware-agnostic routing intents called *Flow Objectives* (e.g., policy constraints or forwarding goals). Concurrently, management interfaces such as gNMI (gRPC Network Management Interface) and gNOI (gRPC Network Operations Interface) are exposed on the Northbound interface, enabling control applications to query dynamic network state (e.g., topology graphs, link statuses) and execute operational lifecycle commands (e.g., rebooting or upgrading devices).

Below the control abstraction is the separation of the control plane and the data plane. The Network OS compiles high-level Flow Objectives into low-level, device-specific *Flow Rules*—concrete match-action entries targeted at explicit hardware table from the entire hardware pipeline. These rules are pushed down via gRPC using southbound protocols such as P4Runtime, OpenFlow, or gNMI. The local Switch OS (e.g., Stratum, originally developed by Google) receives these rules and programs the corresponding entries into the hardware switch silicon⁴. Popular commercial implementations of this merchant silicon include Intel’s Barefoot Tofino and Broadcom’s Tomahawk chips.

B. Network-Wide perspective: The Entire Network Data Path

The previous subsection examined the SDN architecture through the lens of a single switch, a single controller, and its Network OS. This subsection zooms out to trace the data path end-to-end: from its origination at an end host, across one or more SDN-enabled switches, to its destination at another end host. As noted earlier, each switch along this path runs its own instance of a Switch OS, functioning as an on-host agent that programs flow rules onto the underlying silicon. The remainder of this subsection focuses on the end host itself, as the origination and termination point of the path; the inter-switch portion of the path—how multiple Switch OS instances are stitched together into a fabric under a common Network OS—is treated separately in Section V and the discussion of the Leaf-Spine Fabric.

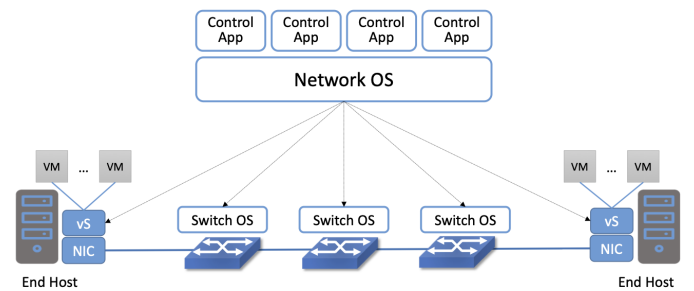


Fig. 3. End-End network view of SDN architecture. Adapted from [1].

⁴Switch silicon refers to a specialized integrated circuit (ASIC) designed for line-rate packet processing, switching, and hardware-accelerated QoS management.

With respect to the end host, Figure 3 highlights three components: the NIC, the virtual switch, and the VMs, all residing on the same physical server. A conventional NIC (Network Interface Card) is a hardware component that connects a computer or device to the network, converting data between the host’s internal representation and the electrical or radio-frequency signals carried over the physical medium, and vice versa. The virtual switch (vSwitch, e.g., Open vSwitch) is a software switch that runs within the server’s hypervisor and binds to the physical NIC as its uplink. Architecturally, the vSwitch occupies the same role as the Switch OS described in the single-switch view: it can be programmed by the same Network OS over southbound protocols such as OpenFlow or P4Runtime, receiving and installing Flow Rules exactly as a hardware Switch OS would. The distinction is one of substrate rather than function—the vSwitch executes its match-action pipeline in software on the host’s general-purpose CPU [2], whereas a physical switch executes the equivalent pipeline on dedicated ASIC silicon. Once programmed, it behaves as an ordinary switch, receiving packets on an input port and forwarding them to the appropriate output port; the endpoints attached to these virtual ports are typically Virtual Machines (VMs) or containers.

That said, this architecture rests on a few assumptions worth making explicit. First, not every end host runs VMs; the same picture applies to containerized or bare-metal hosts, with the vSwitch’s virtual ports simply terminating at a different type of endpoint. Second, the NIC–vSwitch–VM arrangement in Figure 3 is not the only valid way to structure the end host, but rather one particular perspective. This report, following [1], adopts a *network-centric* view, in which the vSwitch and NIC are treated as an extension of the network, controlled by the same Network OS as the rest of the fabric. A *host-centric* perspective is equally valid: it instead treats packet processing as a function of the host operating system itself, which means an architectural change in who is in control—instead of being managed by the network controller, it is configured directly by the host itself, via kernel-resident frameworks such as eBPF/XDP or user-space frameworks such as DPDK [1]. Third, the conventional NIC introduced above is itself only one option: vendors have a long history of offloading kernel functionality onto the NIC, and this has led to the Smart Network Interface Card (SmartNIC), a NIC augmented with an FPGA or ASIC capable of running programmable logic. A SmartNIC can replace the conventional NIC, and, going further, can assist or even replace the vSwitch entirely by implementing part or all of its forwarding pipeline directly in NIC hardware [1]. A more detailed treatment of these host-centric and SmartNIC-based alternatives, including their performance tradeoffs, is left for future work (Section VIII), as it requires background this report has not yet developed.

III. PROGRAMMABLE DATA PLANE AND P4 LANGUAGE

Before turning to what P4 is and how the language itself is structured, it is worth establishing why a new data-plane programming model was needed at all. OpenFlow was built

to reuse capability that switch vendors already had on-chip: McKeown et al. note that “we exploit the fact that most modern Ethernet switches and routers contain flow-tables (typically built from TCAMs)” used internally to implement functions such as firewalling, NAT, QoS, and statistics collection, and that OpenFlow simply exposes “this common set of functions” through a standard interface [4]. This was a deliberate compromise: rather than requiring new switch hardware, the design goal was a datapath “amenable to high-performance and low-cost implementations,” which meant “forgoing the ability to specify arbitrary handling of each packet and seeking a more limited, but still useful, range of actions” [4]. In its original 2008 form this amounted to a single flow table matched against about a dozen header fields such as MAC address, IP address, protocol, and TCP/UDP ports [5].

A. The fundamental limitation of OpenFlow

That original dozen-field flow table did not remain fixed for long. The formal OpenFlow 1.0 specification followed the 2008 proposal in December 2009, and each successive revision of the OpenFlow 1.x line extended the matchable field set to accommodate new headers, so that the number of supported fields rose from 12 to 41 by 2013 [5]. Yet as data-center networking continued to evolve, operators kept demanding support for still newer encapsulation headers (e.g., VXLAN, NVGRE); rather than continuing to extend OpenFlow 1.x indefinitely, researchers proposed a strawman “OpenFlow 2.0” interface instead [5].

The push toward an OpenFlow 2.0 interface reflected two distinct pressures on any successor design: the need to keep pace with the rapid rate of change in networking requirements, and the need for a design that remained viable over the long term. The immediate problem was one of deployment speed. At the time, bare-metal switches still relied on a fixed-function pipeline hardcoded into the switch firmware to match packets, so supporting a new protocol required two slow steps in sequence. First, the protocol itself had to be agreed upon. Second, once agreed upon, the switch firmware itself had to be updated, since the firmware had no built-in intelligence to parse a raw packet format it had never been told about, and changing the firmware took still more time. Consequently, simply adding a new protocol to the specification was not enough on its own: it could still take vendors years to release a new switch model incorporating it, and for that hardware to then be adopted into the already-deployed network. The implication was that future switches should be able to define custom headers themselves, and that controller applications should likewise be able to understand these custom headers.

B. Existing efforts for programmable pipelines

Programmable forwarding pipelines existed well before P4, in several forms, each trading away something essential. Software switches and modular frameworks such as Click [7] offered complete flexibility, but executed on general-purpose CPUs and therefore operated far below the line rates expected of a switch ASIC; Click was also not designed for the

controller-switch split that SDN assumes, and so provided no notion of match-action tables populated at runtime by a controller [5]. At the other extreme, network processors, FPGAs, and—by 2013—reconfigurable match-action ASICs such as RMT [6] had demonstrated that this flexibility could in fact be achieved at terabit speeds. The obstacle there was not capability but usability: each chip exposed its own proprietary low-level interface, which Bosshart et al. compare to microcode programming, so a forwarding behavior written for one target could not be carried to another [5]. The gap P4 was created to fill is therefore a linguistic one rather than a hardware one: it provides a high-level, protocol-agnostic language that a compiler can translate into the configuration each target requires, including the line-rate programmable ASIC pipelines these chips had already made possible [5].

C. Forwarding pipeline

At the architecture level, researchers introduced PISA (Protocol Independent Switching Architecture) as the pipeline design that gives P4 its reconfigurability [1]. As shown in Fig. 4, a PISA pipeline is built from three components—a programmable parser, a programmable multi-stage match-action pipeline, and a programmable deparser—each configured by the P4 program itself rather than hardcoded into switch firmware. This generic, configurable structure is what lets the same abstract model describe packet processing across otherwise substantially different forwarding devices—Ethernet switches, load-balancers, and routers—and across different underlying technologies, whether a fixed-function switch ASIC, an NPU, a reconfigurable switch, a software switch, or an FPGA [5].

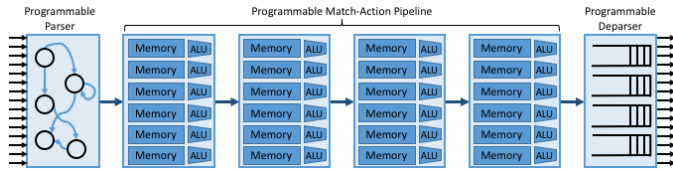


Fig. 4. Abstract forwarding pipeline that model how most data plane devices are forwarding packet. Adapted from [1].

The parser forms the first stage of the pipeline and extracts header fields from an incoming packet by walking a program-defined parse graph: it sequentially extracts each header and copies its values into the memory defined by that header's structure (e.g., the IPv4 header). Once extracted, these fields are passed to the match-action pipeline, where the design's protocol independence is most apparent. Each stage of the pipeline pairs a block of memory—combining SRAM and TCAM [1]—with an ALU. This memory is not preloaded by the switch vendor to recognize a fixed set of headers; that is, it does not inherently support IPv4 but not IPv6, or TCP but not UDP. It instead behaves as a generic bank of lookup-table entries that the P4 program itself populates at configuration time, using the match keys that a given header requires. The lookup is performed by the memory: the

TCAM supports wildcard matching of the arriving packet's fields against the programmed entries, while exact matches are served from SRAM at lower cost [1]. Only after a match is found does the ALU execute the action associated with the matched entry. Because each stage can be pipelined—one stage processing one packet while the next stage concurrently processes another—this reprogrammable design still sustains line-rate throughput [1]. Finally, the deparser reassembles the modified header fields into a single serialized packet before it leaves the pipeline, following the header layout defined by the program.

D. Abstracting the pipeline

E. Example of a P4 program

The preceding subsections described the forwarding pipeline in the abstract; this subsection grounds that description in a concrete program. The example used throughout is `main.p4`, the P4 program written for this project, which is compiled for and executed on BMv2, the reference P4 software switch [10]. The program declares a range of header formats—Ethernet, IPv4, IPv6, TCP, UDP, ICMP, ICMPv6, and NDP—together with Segment Routing over IPv6 (SRv6) headers and a pair of CPU headers that carry packet-in and packet-out messages between the pipeline and the controller. The subsections that follow examine the program in the order a packet encounters it: the architecture it must conform to, the header formats it defines, the parser that extracts them, the tables that match on the extracted fields, and the actions those tables invoke.

1) *The V1Switch pipeline:* BMv2 executes programs written for the `v1model` architecture [10], whose `V1Switch` package fixes both the blocks a program must provide and the order in which they run: a parser, a checksum-verification block, an ingress pipeline, an egress pipeline, a checksum-computation block, and a deparser (Listing 1). A program cannot add, reorder, or omit a stage; a block it does not need must still be supplied with an empty body. These six are only the programmable stages. As shown in Fig. 5, the architecture also places a fixed-function traffic manager between the ingress and egress pipelines, which queues, replicates, and schedules packets [1]; it is not expressed in P4 and is directed only indirectly, through the metadata written during ingress.

```
V1Switch(
    ParserImpl(),
    VerifyChecksumImpl(),
    IngressPipeImpl(),
    EgressPipeImpl(),
    ComputeChecksumImpl(),
    DeparserImpl()
) main;
```

Listing 1. Instantiating the `V1Switch` pipeline in `main.p4`.

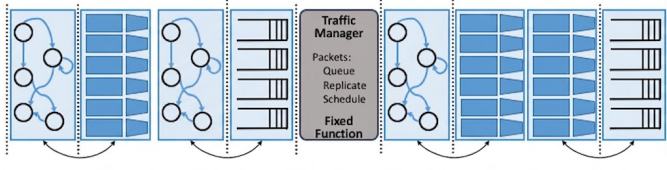


Fig. 5. Programmable pipelines separated by a fixed-function traffic manager. Adapted from [1].

2) *Header and metadata formats*: P4 uses a C-like syntax for its type declarations, which makes this part of the program largely self-explanatory. A header is declared as an ordered sequence of named fields with explicit bit widths, and the program is free to define any header it requires; Listing 2 shows the Ethernet and IPv4 definitions. Widths may be given names of their own through `typedef`—for example `typedef bit<48> mac_addr_t`, naming the 48 bits that hold a MAC address. Values the program matches against are declared as named constants, such as `IP_PROTO_ICMP = 1` and `IP_PROTO_TCP = 6`, whose values are the protocol numbers assigned in RFC 790 [8]. Alongside the headers, the program declares a metadata structure carrying the per-packet state the pipeline needs but that is not necessary to appear on the wire.

```
#include <core.p4>
#include <v1model.p4>

typedef bit<48> mac_addr_t;
typedef bit<32> ipv4_addr_t;

const bit<16> ETHERTYPE_IPV4 = 0x0800;
const bit<8> IP_PROTO_ICMP = 1;
const bit<8> IP_PROTO_TCP = 6;

header ethernet_t {
  mac_addr_t dst_addr;
  mac_addr_t src_addr;
  bit<16> ether_type;
}

header ipv4_t {
  bit<4> version;
  bit<4> ihl;
  bit<6> dscp;
  bit<2> ecn;
  bit<16> total_len;
  bit<16> identification;
  bit<3> flags;
  bit<13> frag_offset;
  bit<8> ttl;
  bit<8> protocol;
  bit<16> hdr_checksum;
  bit<32> src_addr;
  bit<32> dst_addr;
}
```

Listing 2. Header definitions in `main.p4`.

3) *Packet parser*: The parser is expressed as a finite state machine. Each state is introduced by the `state` keyword, and the transition out of that state is declared inside the state body itself.

```
parser ParserImpl (packet_in packet,
                  out parsed_headers_t hdr,
                  inout local_metadata_t l_metadata,
                  inout standard_metadata_t metadata) {

  state start {
    transition select (metadata.ingress_port) {
      CPU_PORT: parse_packet_out;
      default: parse_ethernet;
    }
  }

  state parse_packet_out {
    packet.extract(hdr.cpu_out);
    transition parse_ethernet;
  }

  state parse_ethernet {
    packet.extract(hdr.ethernet);
    transition select (hdr.ethernet.ether_type) {
      ETHERTYPE_IPV4: parse_ipv4;
      ETHERTYPE_IPV6: parse_ipv6;
      default: accept;
    }
  }
}
```

Listing 3. Parser state machine in `main.p4`.

Execution begins in the built-in `start` state; each subsequent state extracts one header from the packet into the header structure declared earlier, then transitions to the state responsible for interpreting the bytes that follow. The machine terminates in one of two built-in states: `accept`, marking the end of parsing and handing the extracted headers to the pipeline, or `reject`, marking a parsing failure [9].

4) *Ingress processing*:

5) *Egress processing*:

6) *Deparser*:

F. Compiling a P4 program

IV. SWITCH OS

V. NETWORK OS

VI. LEAF-SPINE FABRIC

VII. PUTTING IT TOGETHER: THE FULL FABRIC

VIII. RESEARCH DIRECTION

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